

Physik II — FS 2024 — Prof. R. Wallny — Serie 12

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Exercise 1: Self-induction and Spark-plug

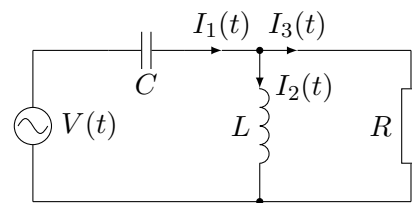
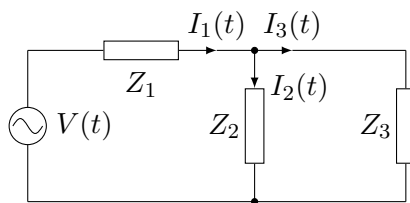
Objective: Applications of self-inductance

Try to answer the following two questions related to induction:

- Why should one never turn electrical equipment on *or* off at the smell of gas in a room?
- How does a spark-plug in a car work? Try to come up with a circuit that could act as a spark-plug.

Exercise 2: Resonance curve

Objective: Impedance, phase difference, resonance.



- Consider the left circuit with the impedances Z_1 , Z_2 and Z_3 , and a generator that provides the complex voltage $V(t) = V_0 e^{i\omega t}$. Calculate the complex current $I_3(t)$ flowing through the impedance Z_3 .
- Consider now the right circuit. Determine the modulus $|I_3|$ and the phase difference α of the complex current $I_3(t) = |I_3| e^{i(\omega t + \alpha)}$ that flows through the resistor in this case. Furthermore, sketch $|I_3|$ as a function of $\omega \in [0, \infty[$ in case of a large resistor R .

Hint: For simplifying the term describing the phase difference, start by deriving a general term for the argument of the reciprocal of a complex number $\arg(1/z)$.

Exercise 3: Disc in a magnetic field

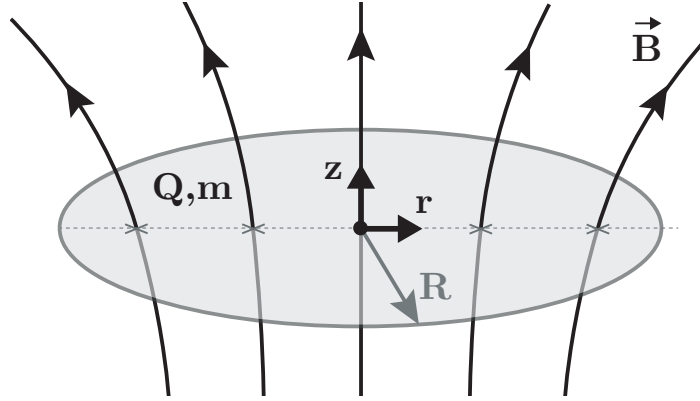
Objective: Faraday's law

Consider a non-conductive disc in the xy plane (at $z = 0$), with radius R and mass m . The disc is uniformly charged with total charge Q .

At time $t = 0$ a magnetic field with cylindrical symmetry is turned on. The components of the field for $t > 0$ are as follows:

$$\vec{\mathbf{B}}(r, \phi, z) = \begin{pmatrix} B_r \\ B_\phi \\ B_z \end{pmatrix} = \begin{pmatrix} rB_0t \\ 0 \\ (k - 2z)B_0t \end{pmatrix} \quad (1)$$

Note that k is a constant, positive length and B_0 is a positive constant with dimensions $[B_0] = \text{m}^{-1}\text{s}^{-1}\text{T}$. For $t < 0$, the magnetic field is zero $\vec{\mathbf{B}} = \vec{\mathbf{0}}$.



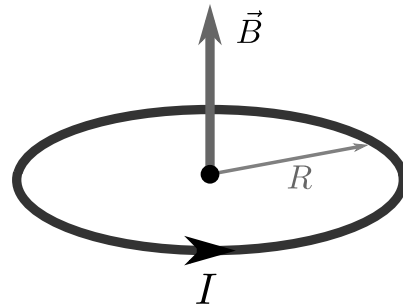
- a) Calculate the magnetic flux at time t (with $t > 0$) through the disc.
- b) Calculate the electric field $\vec{\mathbf{E}}_{\text{ind}}$ for $z = 0$ as a function of the radial distance r from the center of the disc and the time t . Assume that the field for $z = 0$ has the form:

$$\vec{\mathbf{E}}_{\text{ind}}(r, \phi, z = 0) = \begin{pmatrix} E_r \\ E_\phi \\ E_z \end{pmatrix} = \begin{pmatrix} 0 \\ E(r, t) \\ 0 \end{pmatrix} \quad (2)$$

- c) What is the voltage difference between $r = 0$ and $r = R$ due to the presence of $\vec{\mathbf{E}}_{\text{ind}}$?
- d) Using the result from (b) compute the torque $\vec{\mathbf{M}}$ acting on the disc.
Hint: $\vec{\mathbf{M}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$
- e) Assuming that the disc is motionless at $t = 0$, find the angular velocity of the disc at a later time $t > 0$.
Hint: The moment of inertia of the disc around its z axis is $\frac{1}{2}mR^2$.
- f) What is the magnetic field $\vec{\mathbf{B}}_s$ in the disc center ($r = 0, z = 0$) produced by the rotation of the disc?

The magnetic field in the center of a circular current loop with radius R (see the figure on the right) is given by:

$$\vec{\mathbf{B}}_k = \frac{\mu_0 I}{2R} \vec{\mathbf{e}}_z$$



Exercise 4: Maxwell's displacement current

Objective: Maxwell's equations, displacement current

One of the Maxwell's equations reads

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right),$$

where the second term on the right side is called 'Maxwell's displacement current', which - as opposed to $\vec{J}$ - does not require an electrical conductor.

$$\vec{B} = \frac{\mu_0 V}{2\pi R} e^{-t/RC} \left(\frac{r}{r_0^2} \right) \hat{\phi}$$

is the magnetic field inside of a circular parallel-plate capacitor, where V is the applied voltage, R the resistance of the wires and r_0 the radius of the circular plates. Use the Maxwell-equation to calculate the displacement current. In which direction will this current be pointed?

Hint: Use the rotation of a vectorial field $\vec{F}$ in cylindrical coordinates.

Exercise 5: Maxwell's equations and EM waves

Objective: Understanding Maxwell's equations

The Maxwell equations take the following form in vacuum and the absence of any sources :

$$\begin{aligned} \vec{\nabla} \cdot \vec{E} &= 0 & \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \cdot \vec{B} &= 0 & \vec{\nabla} \times \vec{B} &= \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

This is a system of coupled differential equations of first order. The electromagnetic waves are a solution to this system of equations (see script). Consider now an electric field

$\vec{E}(x, t) = E_0 \cos(kx - \omega t) \vec{e}_y$ in the vacuum, where $\vec{e}_y$ is the unit vector in y -direction.

a) Find the corresponding magnetic field $\vec{B}$.

Hint: Start with the general wavefunction for the $\vec{B}$ -field and deduce the unknown parameters by inserting it into the Maxwell equations.

b) Find the correlation between the phase velocity of the wave $v = \frac{\omega}{k} = c$ and the dielectric constant ϵ_0 in vacuum and the magnetic permeability in vacuum μ_0 . Which measurements can be made based on this equation?

c) Sketch $\vec{E}$ and $\vec{B}$ for $t = 0$ as a function of x .

d) The linearity of Maxwells differential equations allows for the superposition of the solutions, i.e. if $\vec{E}$ and $\vec{E}'$ are solutions of the equations, then also $\vec{E}^{\text{tot}} = \vec{E} + \vec{E}'$ is a solution.

Let us assume, that an additional wave $\vec{E}' = \vec{E}_0 \cos(kx - \omega t + \phi) \vec{e}_z$ is added to the original $\vec{E}$ -field. Show that for a fixed point in space the total field $\vec{E}^{\text{tot}}$ describes a circle in the yz -plane for $\phi = 90^\circ$. What happens for $\phi = 0$?